

Green subsidies with demand distortions

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Pigou: Correct externalities at the margin

- ▶ Optimal policy: tax (subsidy) equal to the marginal externality cost (benefit)
Pigou (1920)
- ▶ Examples: gasoline tax
- ▶ Corrects all relevant margins: selection, treatment
- ▶ Has inspired many programs globally subsidizing clean technology usage costs

Principals use both MC and FC green technology subsidies

Marginal cost subsidy

Fixed cost subsidy

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Fixed cost subsidy

e.g. EV charging, electric cooking:

Electric Vehicle (EV) rate plans

Making sense of EV rate plans

Residential EV rates

You can lower your energy costs by enrolling in one of our EV rate plans:

- Home Charging EV2-A
- Electric Vehicle Rate Plan EV-B

Volvo Offers Swedish Buyers A Year Of Free Charging To Spark EV Sales

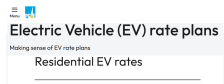
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THURSDAY, DECEMBER 26, 2024 — UPDATED ON MAY 28, 2025 — 5

UMEME Customers Enjoy Cheaper Cooking Tariff Set By ERA

Christopher Kiiza · November 14, 2022 3 minutes read

eCooking Tariff Development

- Test the viability and impact of a specialized eCooking tariff for Tier 3+ connected households in Kenya

Cost: USD 450,000
(KES 57 million)

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UP TO

\$2,000

IN TAX CREDITS FOR HEAT PUMP¹ INSTALLATIONS



Heat pump grant increased to £7,500 by government in efforts to lower pollution and bills

OR UP TO

\$2,600

IF INSTALLING A QUALIFYING HEAT PUMP AND GAS FURNACE SYSTEM

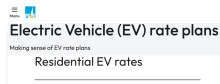


Here are the cars eligible for the \$7,500 EV tax credit in the Inflation Reduction Act in 2023

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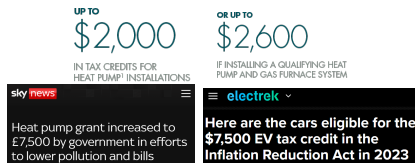
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Tax waivers on eCooking

- A dedicated study to define a suitable tax waiver regime to reduce upfront appliances costs to locally manufactured appliances, components and/or accessories, which will increase demand.
- Determine the economic and financial impact of these waivers on government revenues and household affordability.

Cost: USD 125,000
(KES 16 Million)



PROJECT TYPE: Biogas
LOCATION: Kenya
VINTAGES: 2023

SCALE: Large

Market distortions: $\text{marginal benefit} \neq \text{marginal cost}$

- ▶ Market distortions are pervasive

Atkin et al (2025); Bergquist, Lashkari, Verhoogen (2026); Ghatak and Mookherjee (2025)

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- ▶ Inattention to benefits *Dellavigna (2009)*
- ▶ Present bias *Laibson (1997); O'Donoghue, Rabin (1999); Indarte, Kluender, Malmendier, Stepner (2026)*
- ▶ Credit constraints, caused by informality, imperfect property rights, institutional constraints, judicial frictions, etc. [Detail](#)

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- ▶ Theory of the second best: Pigouvian intuition may not hold in the presence of additional distortions *Lipsey & Lancaster (1956)*

This paper (1) In theory, demand distortions can **increase** efficacy of fixed cost subsidies

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Yes! Distortions can:

- ▶ Increase the efficacy of fixed cost subsidies relative to marginal cost subsidies
- ▶ Increase abatement per subsidy dollar in absolute terms

Energy-intensive durable: Stoves

- ▶ Billions of people cook with biomass every day (usually wood or charcoal)
- ▶ This is usually a household's most polluting device: one charcoal stove emits approx. **4.9 tCO₂e** per year
- ▶ For comparison: driving a gasoline vehicle 11,500 miles at 22.2 MPG emits **4.6 tCO₂** per year (EPA, 2025)



This paper (2) Impact of induction stove subsidies in Kenya



Kenyan Jiko



Induction stove

Randomized Controlled Trial

1. **Sample:** 2,134 Kenyan households
2. **Randomized treatments:**
 - ▶ **Marginal cost subsidies:** 0, 25%, or 75% off the cost of electricity used to cook with the induction stove
 - ▶ **Fixed cost subsidies:** 10% or 75% off the standard price of \$82
 - ▶ **Payment structure:** pay up front or pay with loan
3. **Measurement:**
 - ▶ 2min-frequency temperature monitors on charcoal stoves for 3+ months
 - ▶ 15min-frequency induction stove usage for 3+ months
 - ▶ In-person 1-month follow up survey

Preview: Impacts

1. The induction stove reduces monthly energy spending by \$8.9
 - ▶ \$184 over 1.9-year lifespan; 112% ROI on \$82 price
 - ▶ Monthly charcoal spending down \$10 (66%)
 - ▶ Monthly electricity spending up \$2.3 (65%)
2. Net abatement of 2.5 tCO₂e per year
 - ▶ For comparison: an EV abates between 0.1–1.5 tCO₂e per year
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 - ▶ Without a loan, average willingness-to-pay is \$20: $\bar{\gamma} = 0.11$
 - ▶ With a loan, average willingness-to-pay is \$35: $\bar{\gamma} = 0.19$

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4. Marginal cost subsidies have no impacts on adoption or usage

Preview: How demand distortions affect subsidy efficacy

	Less distorted (loan)	More distorted (cash)	Difference
Observed willingness-to-pay	\$35	\$20	-45%

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<i>Channel 1: Marginal adopter</i>			
Fuel savings per marginal adoption	\$154	\$200	+30%
Abatement per marginal adoption	4.2 tCO ₂ e	5 tCO ₂ e	+19%
<i>Channel 2: Subsidy cost</i>			
Subsidy spending per marginal adoption	\$93	\$65	−30%

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<i>Aggregate difference</i>			
Observable welfare gain per \$1 spending	\$6.1	\$11	+78%
Subsidy spending per tCO ₂ e abated	\$22	\$13	−41%

Preview: Model estimation and counterfactuals

- ▶ Our context still has significant demand distortions:
even with a loan, we estimate $wtp_{observed} = 0.19 \cdot wtp_{efficient}$
- ▶ What if demand distortions were reduced further?

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even with a loan, we estimate $wtp_{observed} = 0.19 \cdot wtp_{efficient}$
- ▶ What if demand distortions were reduced further?
- ▶ We use 2-step Generalized Method of Moments (GMM) to estimate 5 model parameters by matching to 7 moments observed in the experiment
- ▶ Counterfactual subsidy spending under $wtp_{observed} = \bar{\gamma} \cdot wtp_{efficient}$:
 - ▶ At $\bar{\gamma} = 0.1$, \$13 per ton of CO₂ abated
 - ▶ At $\bar{\gamma} = 0.5$, \$68 per ton of CO₂ abated
 - ▶ At $\bar{\gamma} = 1$, \$137 per ton of CO₂ abated

Efficient carbon abatement at scale

- ▶ Annual carbon mitigation financing exceeds \$1 trillion *Buchner, 2024*
- ▶ Concerns about adverse selection and moral hazard
Einav, Finkelstein, and Cullen, 2010; Jack, Jayachandran, Kala, Pande, 2025; Aspelund and Russo, 2026; Chen, Ryan, and Xu, 2026
- ▶ Efficiency: Allocate marginal dollars to lowest cost abatement technologies

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 - ▶ Efficiency: Allocate marginal dollars to lowest cost abatement technologies
- Green subsidy expenditure could abate more in markets with larger frictions
- ▶ Weak institutions in many low- and middle-income countries
⇒ global capital misallocation
Acemoglu, Johnson, and Robinson, 2005; Nunn, 2007; Shapiro, 2025
 - ▶ U.S.: average APR 6% ⇒ \$1,300 per tCO₂e
Experian, 2025, Hahn et al., 2026
 - ▶ Kenya: APR 224% (even with digital collateral) ⇒ \$13 per tCO₂e

Interest rates

Today

Background

Theory

RCT Design

Impact

Externality of the marginal adopter

Subsidy expenditure per marginal adoption

Model estimation and counterfactuals

Implications for abatement policy

First, a simplified model to build mechanisms intuition

- ▶ Each individual has inelastic energy services demand θ_i (e.g. cooking, driving)
- ▶ They can satisfy this demand with a dirty or greener version of a technology
- ▶ Benefits of clean energy:

$$\underbrace{\pi_i \equiv \theta_i(p_H - p_L)}_{\text{private (fuel savings)}}$$

$$\underbrace{\phi_i \equiv \theta_i(\bar{\phi}_H - \bar{\phi}_L)}_{\text{social (externality reduction)}}$$

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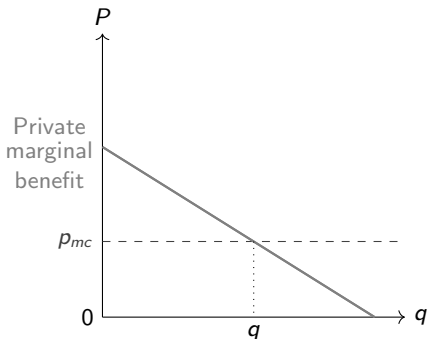
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1. $wtp_i = \gamma \pi_i$: We focus on $\gamma < 1$, e.g. credit constraints
2. Market demand: $Q(p - s_1) = \sum_i \mathbb{1}[wtp_i \geq p - s_1]$

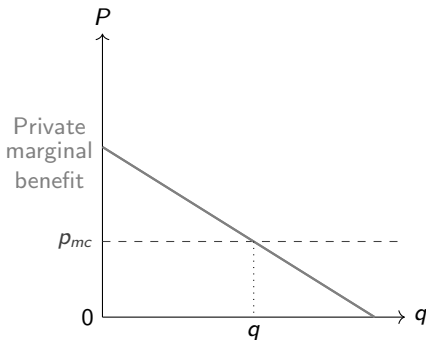
Distortions can increase benefit per **fixed cost** subsidy (s_1)

(a) No distortion ($\gamma = 1$)

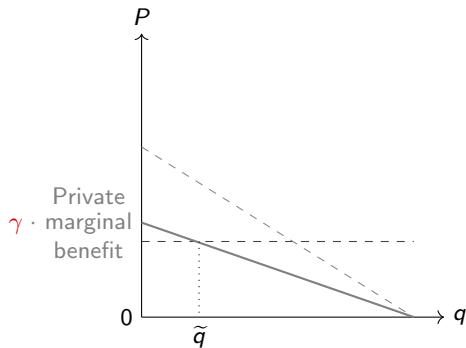


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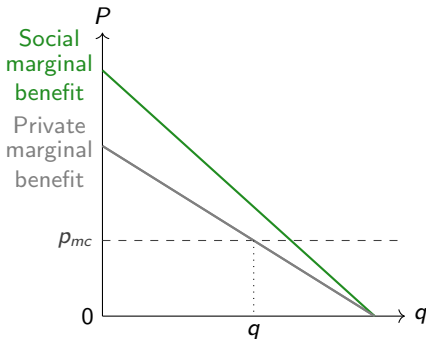


(b) Demand distortion ($\gamma < 1$)

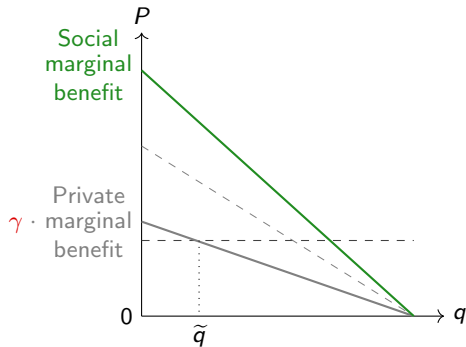


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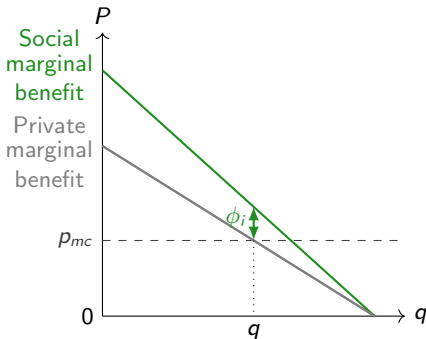
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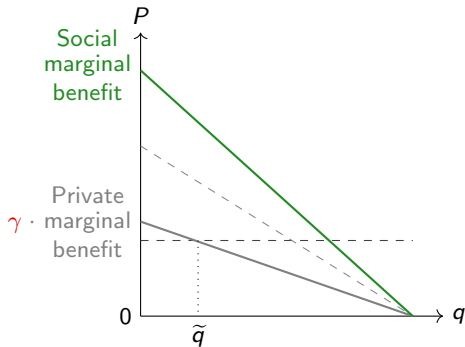
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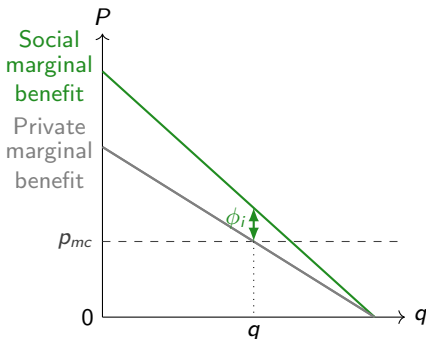
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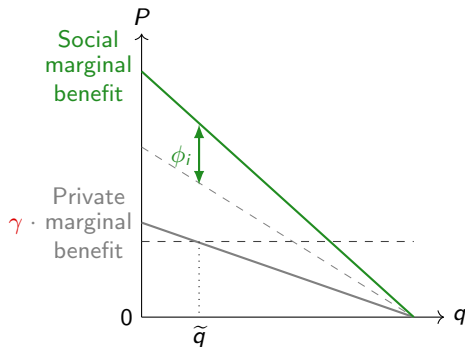
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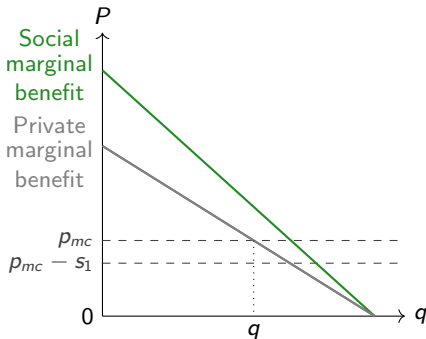
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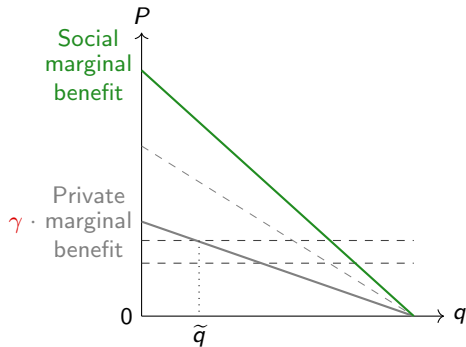
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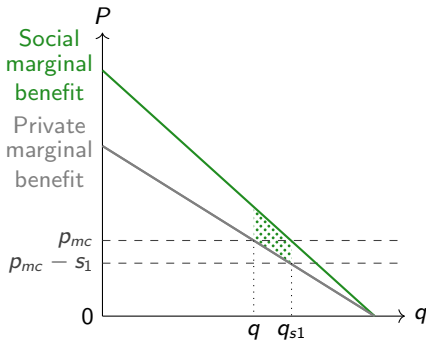
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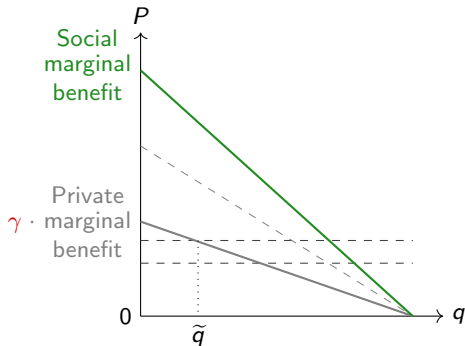
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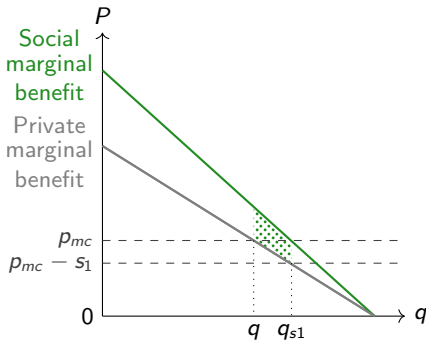


Environmental benefit

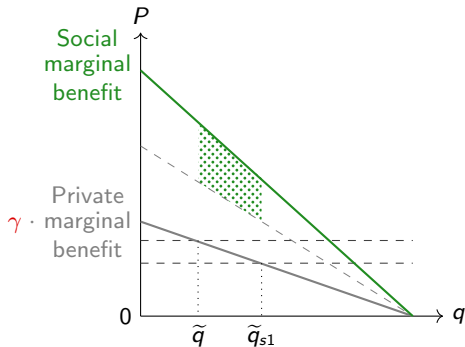
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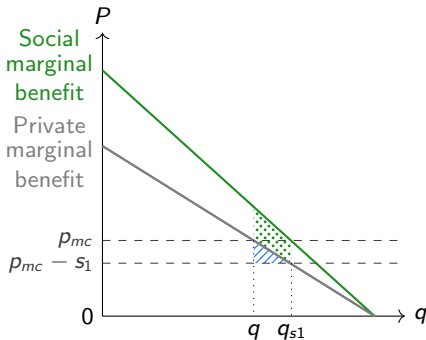


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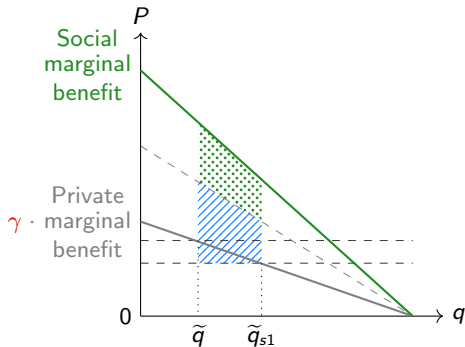
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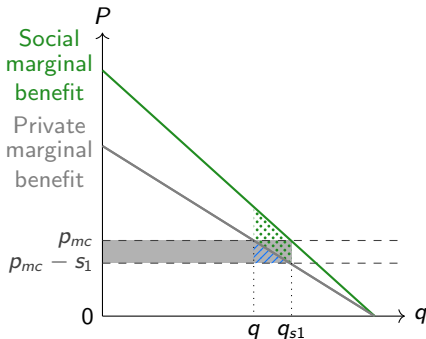
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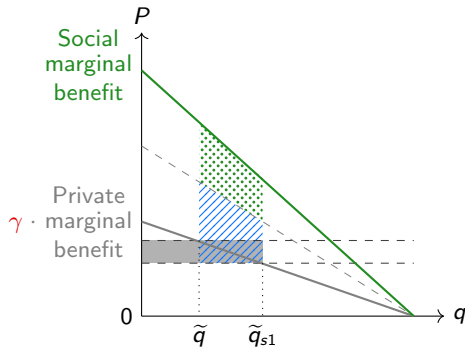
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Channel 2: More elastic demand reduces cost per marginal adopter

(a) No distortion ($\gamma = 1$)



(b) Demand distortion ($\gamma < 1$)



Environmental benefit

Consumer surplus

Subsidy cost

Full model: Set of policy instruments

Principal sets three policies:

1. **Fixed cost subsidy** s_1 : purchase price discount
2. **Marginal cost subsidy** s_2 : electricity price reduction
3. **Marginal cost tax** τ : fuel price increase

Full model: Adoption and usage

Full model

- ▶ Individual preference parameter α (e.g. ease, joy of using electric)
- ▶ Uniform demand distortions along two dimensions:
 1. Demand dampened by γ - affects demand but not usage
 2. Perceived private benefits dampened by δ - also affects usage [not today]
- ▶ Agents make two decisions:
 1. **Adopt**: buy the green technology at subsidized price
 2. **Usage**: fraction of energy services via green tech, generates usage benefit
- ▶ Throughout we assume standard bounds on derivatives and elasticities
Allcott, Mullainathan, Taubinsky (2014)

Optimal policy: Social planner

- ▶ **Objective:** maximize total welfare
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- ▶ **With distortion** ($\gamma < 1$):

$$s_1^* = p(1 - \gamma) > 0, \quad s_2^* + \tau^* = \phi$$

MC tax/subsidy fixes usage, but too few adopt $\Rightarrow$ FC subsidy closes the gap

Optimal policy: Constrained principal

- ▶ **No authority to tax:** can subsidize purchase price (s_1) or usage price (s_2)
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- ▶ **Objective:**

$$(s_1^*, s_2^*) = \arg \max_{\underline{\alpha}} \int_{\underline{\alpha}}^{\infty} V(\alpha) g(\alpha) d\alpha$$

- ▶ **Budget constraint:**

$$\underbrace{s_1 \cdot Q}_{\text{FC subsidy cost}} + \underbrace{s_2 \cdot U}_{\text{MC subsidy cost}} \leq B$$

where Q = number of adopters, U = total clean energy usage

Optimal policy: Constrained principal

$$\tilde{x}^* + \underline{\alpha} = \text{usage}$$

$$\pi = \text{fuel savings}$$

$$\phi = \text{environmental benefit}$$

1. **Social principal:** values private welfare and externalities; evaluate MVPF; 'inframarginal transfers are transfers'
 - ▶ $V(\underline{\alpha}) = (\tilde{x}^* + \underline{\alpha})(\pi + \phi) + \mu(\tilde{x}^*) - p$
 - ▶ Examples: Government agencies
2. **Environmental principal:** maximize CO₂e abatement; 'inframarginal transfers are wasted subsidy expenditures'
 - ▶ $V(\underline{\alpha}) = (\tilde{x}^* + \underline{\alpha})\phi$
 - ▶ Examples: UNFCCC, CORSIA, Article 6.4, carbon credit buyers

Optimal policy: Constrained principal

$\tilde{x}^* + \underline{\alpha} = \text{usage}$

$\pi = \text{fuel savings}$

$\phi = \text{environmental benefit}$

$$V(\underline{\alpha}) = (\tilde{x}^* + \underline{\alpha})(\pi + \phi) + \mu(\tilde{x}^*) - p$$

FOC for fixed cost subsidy s_1 : subsidy affects extensive margin only

$$\lambda = \frac{\overbrace{V(\underline{\alpha})}^{\text{value per new adopter}}}{\underbrace{s_1 + s_2(\tilde{x}^* + \underline{\alpha})}_{\text{total subsidy to new adopters}} + \underbrace{\frac{p - s_1}{|\tilde{\mathcal{E}}(p - s_1)|}}_{\text{inframarginal cost}}}$$

Optimal policy: Constrained principal

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$\phi = \text{environmental benefit}$

$$V(\underline{\alpha}) = (\tilde{x}^* + \underline{\alpha})(\pi + \phi) + \mu(\tilde{x}^*) - p$$

FOC for marginal cost subsidy s_2 : subsidy affects extensive and intensive margins

$$\lambda = \frac{\overbrace{[\pi + \phi + \mu'(\tilde{x}^*)]}^{\text{value of additional usage}} \frac{d\tilde{x}^*}{ds_2} Q + \overbrace{V(\underline{\alpha})g(\underline{\alpha})}^{\text{value of new adopters}} \left(-\frac{\partial \underline{\alpha}}{\partial s_2} \right)}{\underbrace{s_2 \frac{d\tilde{x}^*}{ds_2} Q}_{\text{cost of increased usage}} + \underbrace{[s_1 + s_2(\tilde{x}^* + \underline{\alpha})] g(\underline{\alpha}) \left(-\frac{\partial \underline{\alpha}}{\partial s_2} \right)}_{\text{total subsidy to new adopters}} + \underbrace{U}_{\text{inframarginal cost}}}$$

Optimal policy: Constrained principal

Equalizing the FOC and simplifying:

$$\frac{ds_1^*}{d\gamma} = \underbrace{\left. \frac{\partial s_1}{\partial \gamma} \right|_{s_2}}_{\text{direct effect}} + \underbrace{\left. \frac{\partial s_1}{\partial s_2} \right|_{\gamma}}_{\text{re-optimization}} \cdot \frac{ds_2^*}{d\gamma} < 0$$

In other words, distortions γ change two things:

1. **Direct effect:** Lower γ changes the s_1 required to incentivize a given α_i
 $\Rightarrow$ always negative
2. **Re-optimization:** γ affects s_2^* which affects s_1^* via the budget constraint
 $\Rightarrow$ negative under our assumptions

Optimal policy: Model results

Subsidy efficacy: welfare gain per dollar of subsidy expenditure

- ▶ A demand distortion increases the **subsidy efficacy** of a given fixed cost subsidy

Optimal policy: Model results

Subsidy efficacy: welfare gain per dollar of subsidy expenditure

- ▶ A demand distortion increases the **subsidy efficacy** of a given fixed cost subsidy
- ▶ If the principal can re-optimize between fixed and marginal subsidies then the effect of demand distortion on **subsidy efficacy** depends on parameters

Optimal policy: Model results

Subsidy efficacy: welfare gain per dollar of subsidy expenditure

- ▶ A demand distortion increases the **subsidy efficacy** of a given fixed cost subsidy
- ▶ If the principal can re-optimize between fixed and marginal subsidies then the effect of demand distortion on **subsidy efficacy** depends on parameters
- ▶ Even with re-optimization, demand distortions increase **subsidy efficacy** in many contexts

Today

Background

Theory

RCT Design

Impact

Externality of the marginal adopter

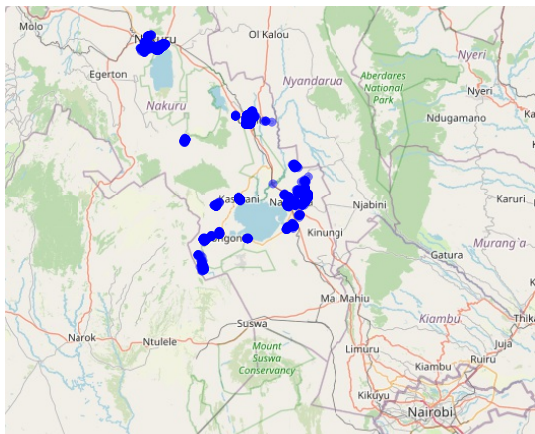
Subsidy expenditure per marginal adoption

Model estimation and counterfactuals

Implications for abatement policy

RCT in Nakuru county with 2,134 households

- ▶ Peri-urban regions around Naivasha, Nakuru, Gilgil
- ▶ 65% of households have a grid electricity connection (*Kenya: 50%*)
- ▶ 69% of households cook with charcoal or firewood (*Kenya: 65%*)



2,134 study participants

Enrolled at their homes by Busara enumerators, subject to eligibility criteria:

- ▶ Use traditional charcoal cookstove as primary cooking technology
- ▶ Spend at least \$4 per week on charcoal (5–10% of household income)
- ▶ Have a pre-pay electricity meter

Study participants

Attrition

Randomly assigned credit, FC subsidies, MC subsidies Balance

- ▶ **Fixed cost subsidies**, down from store price of \$82:
 - ▶ 46% assigned a price of \$20 (75% subsidy)
 - ▶ 49% assigned a price of \$73 (10% subsidy)

Randomly assigned credit, FC subsidies, MC subsidies Balance

- ▶ **Fixed cost subsidies**, down from store price of \$82:
 - ▶ 46% assigned a price of \$20 (75% subsidy)
 - ▶ 49% assigned a price of \$73 (10% subsidy)
- ▶ **Payment structure**: up front or \$12 down payment + 90 day loan

Randomly assigned credit, FC subsidies, MC subsidies Balance

- ▶ **Fixed cost subsidies**, down from store price of \$82:
 - ▶ 46% assigned a price of \$20 (75% subsidy)
 - ▶ 49% assigned a price of \$73 (10% subsidy)
- ▶ **Payment structure**: up front or \$12 down payment + 90 day loan
- ▶ **Marginal cost subsidies**: 0%, 25%, or 75% subsidy on the cost of electricity used by the induction stove for 6 months after purchase
 - ▶ Implementation similar to Kenya Power subsidy program
- ▶ All treatments fully cross-randomized

Information about marginal costs of electricity

[All flyers](#)

Information (85%)

EOCA YANGU ITATUMIA UNIT/TOKENS NGAPI ZA STIMA KUPIKA?

MAKADIRIO KWA FAMILIA YA WATU 4



UGALI

5sh

1sh kwa miezi
6 ya kwanza



SUKUMA WIKI

5sh

1sh kwa miezi
6 ya kwanza



RICE

10sh

3sh kwa miezi
6 ya kwanza



BEANS

25sh

6sh kwa miezi
6 ya kwanza



FISH

5sh

1sh kwa miezi
6 ya kwanza



ecoq
stoves for life

25sh

6sh kwa miezi
6 ya kwanza

GITHERI



5sh

1sh kwa miezi
6 ya kwanza

TEA



5sh

1sh kwa miezi
6 ya kwanza

MBOGA



5sh

1sh kwa miezi
6 ya kwanza

CHAPATI



20sh

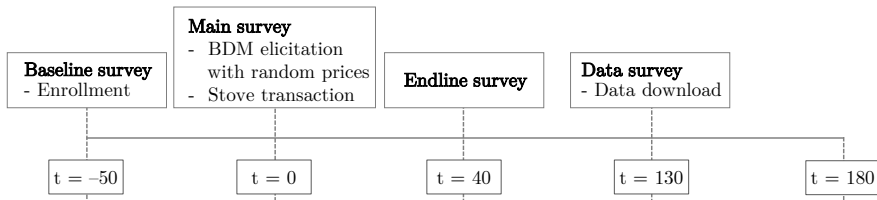
15sh kwa miezi
6 ya kwanza

BEEF STEW



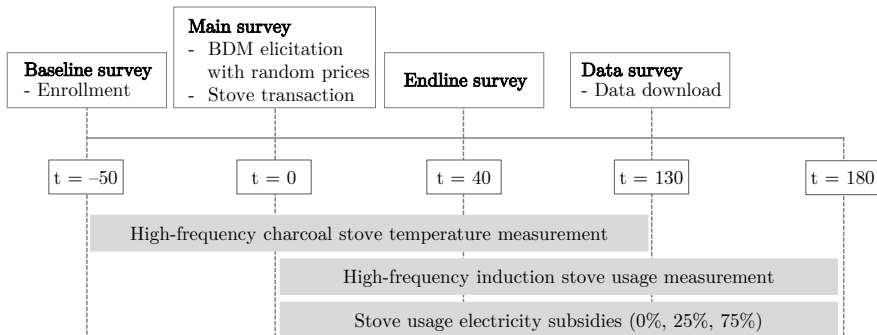
Implementation Timeline

- Implementation: August 2025 – March 2026



Implementation Timeline

► Implementation: August 2025 – March 2026



Main visit timeline

1. Collect Kenya Power meter number
2. Inform about electricity subsidy using informational flyer (if relevant) Flyers
3. Inform how credit works (if relevant)

Main visit timeline

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 - 1) Elicit WTP_i using binary search
 - 2) Reveal random price P_i ; either \$20 or \$73
 - 3) WTP is binding: adopt *iff* $WTP_i \geq P_i$
 - ▶ Pay P_i : incentive compatible
 - ▶ 98.2% of respondents complied

Main visit timeline

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 - ▶ Pay P_i : incentive compatible
 - ▶ 98.2% of respondents complied
5. Buyers: demo stove usage, add to customer service WhatsApp group

Today

Background

Theory

RCT Design

Impact

Externality of the marginal adopter

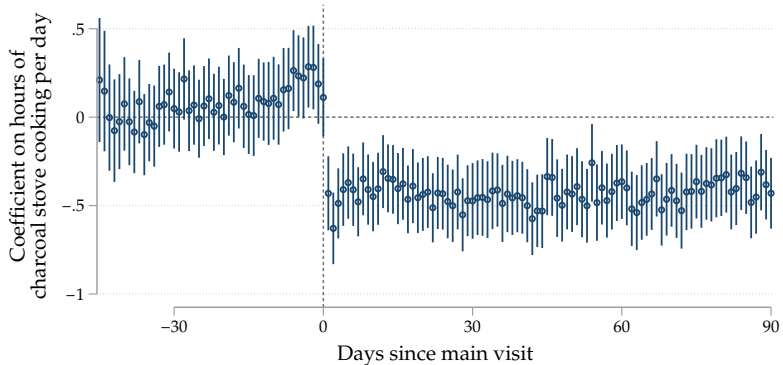
Subsidy expenditure per marginal adoption

Model estimation and counterfactuals

Implications for abatement policy

Results: Induction stove adoption reduces charcoal usage

- ▶ Data source: temperature sensors
- ▶ Control mean: 91 minutes per day
- ▶ OLS treatment effect: -26 minutes per day
- ▶ IV treatment effect: -31 minutes per day



Results: Induction stove adoption reduces energy spending

- Instrumental variables using randomly assigned price as instrument

	Charcoal	LPG	Wood	Electricity	Total
	(1)	(2)	(3)	(4)	(5)
Bought induction stove (=1)	-10.01*** (0.82)	-0.85 (0.56)	-0.33 (0.21)	2.30*** (0.38)	-8.89*** (1.06)
Observations	2081	2081	2081	2081	2081
Control Mean	15.11	5.52	0.74	3.51	24.89
F-Stat	241.34	241.34	241.34	241.34	241.34

- Monthly charcoal spending down \$10 (66%)
- Monthly electricity spending up \$2.3 (65%, incl. subsidy)
- Net savings \$8.9 per month
- \$184 over 1.9-year lifespan (10% discount rate), with an \$82 price: 112% ROI

Lifespan

Result: Induction stove adoption reduces CO₂e emissions

- Instrumental variables using randomly assigned price as instrument

	Charcoal	LPG	Electricity	Total
Bought induction stove (=1)	-2.53*** (0.20)	-0.03 (0.02)	0.02*** (0.00)	-2.53*** (0.20)
Observations	2081	2081	2081	2081
Control Mean	3.77	0.17	0.04	3.99
F-Stat	122	122	122	122

- Average abatement across all study participants over the 1.9-year stove life: 4.8 tCO₂e per stove
- Modest changes in self-reported health symptoms, foods cooked

Three measures of fixed subsidy abatement cost (no loan)

1. **Subsidy cost per tCO₂e**: \$65 in subsidy expenditure per additional adoption, at 5 tCO₂e abatement per adoption ⇒ **\$13 per tCO₂e**
Environmental principal: maximize abatement, minimize subsidy expenditure

Three measures of fixed subsidy abatement cost (no loan)

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2. **Resource cost per tCO₂e**: Stove cost (\$82) – fuel savings (\$200)
⇒ **negative \$24 per tCO₂e**

Social planner: values private savings, disregards inframarginal transfers

Three measures of fixed subsidy abatement cost (no loan)

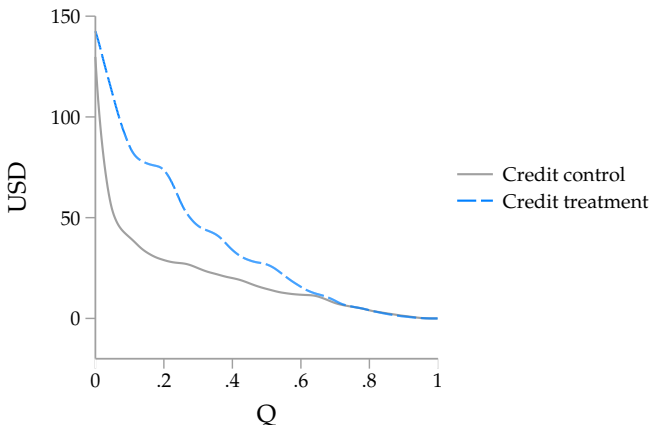
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⇒ **negative \$24 per tCO₂e**
Social planner: values private savings, disregards inframarginal transfers
3. **Observable welfare gain per \$1 spending:** Each \$100 subsidy spending causes 1.5 adoptions—each \$200 fuel savings, \$82 stove cost, 5 tCO₂e
⇒ Only valuing private benefit: \$1.8
⇒ Assuming SCC \$120: \$11
⇒ Assuming SCC \$190–\$380: \$16–\$31

Alternative assumptions

Demand distortions reduce WTP by 45%

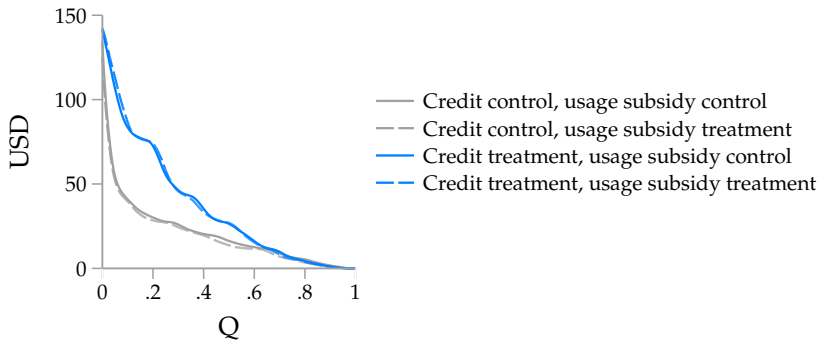
Given \$184 in discounted fuel savings:

- ▶ For participants paying up front, WTP of \$20 implies $\bar{\gamma} = 0.11$
- ▶ For participants paying with loan, WTP of \$35 implies $\bar{\gamma} = 0.19$



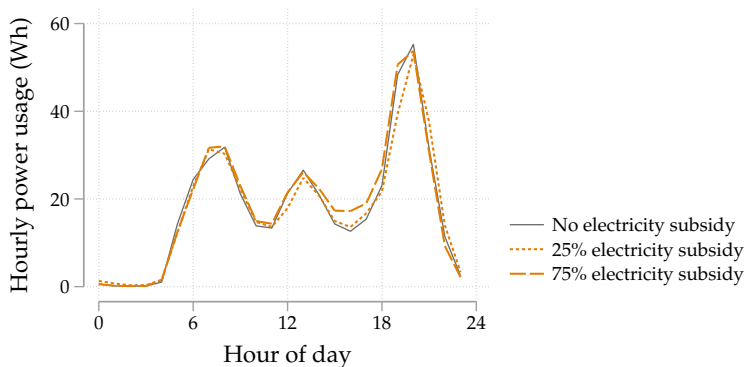
Marginal cost subsidies have no impact on WTP

- ▶ Point estimate: \$0.01 per \$1 subsidy
- ▶ 95% CI: each \$1 of marginal cost subsidy increases WTP by $\leq \$0.16$



Intensive margin is very price inelastic

- ▶ Data source: induction stove 15-minute interval data
- ▶ Cannot distinguish between selection plus direct treatment effect
- ▶ Likely an upper bound on intensive margin elasticity
- ▶ Demand elasticity of stove usage to electricity price: -0.04



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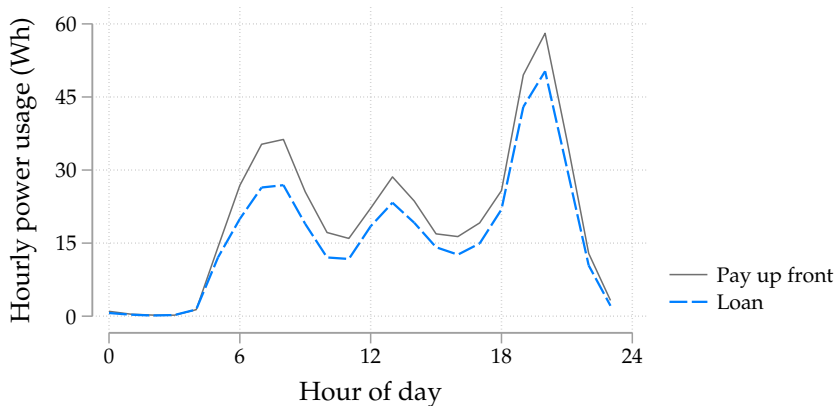
Model estimation and counterfactuals

Implications for abatement policy

Externality of the marginal adopter

- ▶ **Theory:** If demand distortions decrease WTP, and private and social benefits are positively correlated, then distortions increase the positive externality generated by marginal adopters
- ▶ Define marginal adopters as having $WTP \in [20, 73]$
 - ▶ Recall: IV compliers (marginal adopters): $WTP \in [20, 73]$

Marginal adopters with larger distortions use stoves more



Marginal adopters with larger distortions abate more

Charcoal

	(1) USD	(2) USD	(3) CO ₂	(4) CO ₂
Credit (=1)	-1.0 (1.4)	-1.0 (1.4)	-0.2 (0.4)	-0.2 (0.4)
Bought stove (=1)	-10.5*** (0.6)		-2.6*** (0.2)	
Bought X No credit		-10.5*** (0.6)		-2.6*** (0.2)
Bought X Credit	1.6* (0.9)	-8.9*** (0.6)	0.4* (0.2)	-2.2*** (0.2)
Observations	754	754	754	754
Control Mean	15.7	15.7	3.9	3.9

Marginal adopters with larger distortions abate more

	Charcoal				Total			
	(1) USD	(2) USD	(3) CO ₂	(4) CO ₂	(5) USD	(6) USD	(7) CO ₂	(8) CO ₂
Credit (=1)	-1.0 (1.4)	-1.0 (1.4)	-0.2 (0.4)	-0.2 (0.4)	-0.7 (2.0)	-0.7 (2.0)	-0.2 (0.4)	-0.2 (0.4)
Bought stove (=1)	-10.5*** (0.6)		-2.6*** (0.2)		-9.7*** (0.9)		-2.6*** (0.2)	
Bought X No credit		-10.5*** (0.6)		-2.6*** (0.2)		-9.7*** (0.9)		-2.6*** (0.2)
Bought X Credit	1.6* (0.9)	-8.9*** (0.6)	0.4* (0.2)	-2.2*** (0.2)	2.2* (1.2)	-7.4*** (0.8)	0.4* (0.2)	-2.2*** (0.2)
Observations	754	754	754	754	754	754	754	754
Control Mean	15.7	15.7	3.9	3.9	25.5	25.5	4.1	4.1

Summary: Marginal adopters with larger distortions generate larger private and social benefits

	Loan	Pay up front
	$(\bar{\gamma} = 0.19)$	$(\bar{\gamma} = 0.11)$
Monthly fuel savings among marginal adopters	\$7.6	\$10.1
Difference in private benefit	+30%	
Annual tCO ₂ e abated per marginal adopter	2.2 tCO ₂ e	2.6 tCO ₂ e
Difference in marginal externality	+19%	

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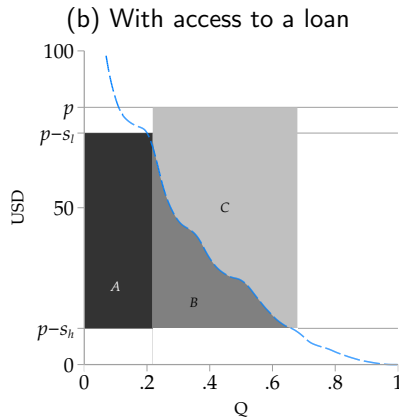
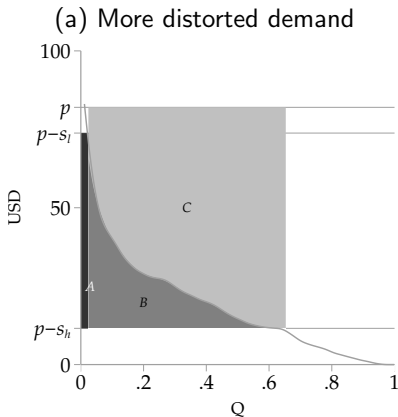
Implications for abatement policy

Demand distortion increases demand elasticity

- ▶ More distorted demand: $\mathcal{E}_D = -2.3$
- ▶ With access to a loan: $\mathcal{E}_D = -0.9$

	Quantity (%)			Log(Quantity)		
	(1)	(2)	(3)	(4)	(5)	(6)
Price (USD)	-0.84*** (0.01)	-0.96*** (0.02)	-0.80*** (0.01)			
Log(Price)				-1.23*** (0.01)	-2.33*** (0.02)	-0.88*** (0.01)
Observations	765	387	378	765	387	378
Sample		C=0	C=1		C=0	C=1

More marginal, fewer inframarginal payments



- (A) Inframarginal payments to inframarginal buyers
- (B) Inframarginal payments to marginal buyers
- (C) Marginal payments to marginal buyers

More marginal, fewer inframarginal payments

	Loan	Pay up front
	$(\bar{\gamma} = 0.19)$	$(\bar{\gamma} = 0.11)$
Extensive margin elasticity to fixed subsidy	$\mathcal{E}_d = -0.9$	$\mathcal{E}_d = -2.3$
Fraction of subsidy expenditures inframarginal	38%	6.5%
Subsidy expenditure per marginal adoption	\$93	\$65
Difference in cost per adoption	-30%	

Summary: Effect of demand distortions on subsidy efficacy

	Less distorted $(\bar{\gamma} = 0.19)$	More distorted $(\bar{\gamma} = 0.11)$	Difference
<i>Channel 1: Marginal adopter</i>			
Fuel savings per marginal adoption	\$154	\$200	+30%
tCO ₂ e abated per marginal adoption	4.2 tCO ₂ e	5 tCO ₂ e	+19%

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Extensive margin elasticity	−0.9	−2.3	
Subsidy spending inframarginal	38%	6.5%	
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<i>Aggregate difference</i>			
Observable welfare gain per \$1 spending	\$6.1	\$11	+78%
Subsidy spending per tCO ₂ e abated	\$22	\$13	-41%

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What if the distortion was even lower?

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Model estimation and counterfactuals

Implications for abatement policy

Model estimation: 2-step GMM

- ▶ Private benefit b_i
 - ▶ Distributed Weibull: $b_i \sim \hat{b} \cdot \text{Weibull}(\Omega_1, \Omega_2)$, $\hat{b} = \$184$
- ▶ Social benefit ϕ_i
 - ▶ Function of private benefit: $\phi_i = \hat{\phi} \cdot \left[\Omega_3(b_i/\hat{b}) + (1 - \Omega_3)\varepsilon_i \right]$, $\hat{\phi} = 4.8 \text{ tCO}_2\text{e}$
 - ▶ Mixing parameter $\Omega_3 \in [0, 1]$ governs correlation between b , ϕ

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 - ▶ Mixing parameter $\Omega_3 \in [0, 1]$ governs correlation between b_i , ϕ_i
- ▶ Heterogeneous demand wedge γ_i
 - ▶ Logit-normal: $\gamma_i = \text{logistic}(\mu_c + \sigma_c \eta_i)$ with $\eta_i \sim N(0, 1)$
 - ▶ Dispersion $\sigma_c = \log(1 + \exp(\Omega_4 + \Omega_5 \bar{\gamma}_c))$
 - ▶ Location μ_c pinned down by $E[\gamma_i|c] = \bar{\gamma}_c$

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- ▶ WTP: $wtp_i = \gamma_i \cdot b_i$

We target 7 empirical moments to estimate the 5 parameters

- ▶ Correlation(b , ϕ) (1)
- ▶ Average marginal externality by randomly assigned payment group (2)
- ▶ Extensive margin elasticity by randomly assigned payment group (2)
- ▶ Cost per marginal adoption by randomly assigned payment group (2)

Model fit

Estimated parameters: $\hat{\Omega}_1 = 1.9$, $\hat{\Omega}_2 = 1$, $\hat{\Omega}_3 = 0.6$, $\hat{\Omega}_4 = -2.7$, $\hat{\Omega}_5 = 27$

Moment	Pay up front ($\bar{\gamma} = 0.11$)			Pay with loan ($\bar{\gamma} = 0.19$)		
	Emp.	Model	Dev.	Emp.	Model	Dev.
Correlation (ρ)	0.8	0.8	-1%	0.8	0.8	-1%
Marginal externality (ϕ)	5	5	2%	4.2	4.2	0%
Extensive margin elasticity ($\mathcal{E}_D$)	-2.3	-2.4	-2%	-0.9	-0.7	15%
USD per marginal adoption	65.4	64.6	-1%	93.4	96.1	3%

Counterfactual I: Varying demand distortion ($\bar{\gamma}$)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Average demand distortion ($\bar{\gamma}$)	0.05	0.11	0.19	0.25	0.5	0.75	1
Marginal externality (ϕ)	7.3	5	4.2	4	3.3	2.9	2.5
Extensive margin elasticity ($\mathcal{E}_D$)	-9.4	-2.4	-0.7	-0.5	-0.2	-0.2	-0.1
USD per marginal adoption	62	65	96	122	223	293	348
<i>Observable welfare gain per subsidy dollar (in USD)</i>							
Observed		11	6				
Simulated	19	11	6	4	2	1	1
<i>Subsidy cost per tCO₂e (in USD)</i>							
Observed		13	22				
Simulated	8	13	23	31	68	102	137

Counterfactual I: Varying demand distortion ($\bar{\gamma}$)

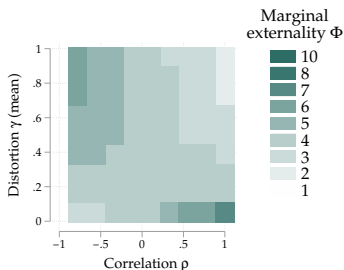
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
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<i>Subsidy cost per tCO₂e (in USD)</i>							
Observed		13	22				
Simulated	8	13	23	31	68	102	137

- Without demand distortions ($\bar{\gamma} = 1$), abatement cost would be $\geq 10\times$ **higher**

Counterfactual II: Varying correlation and distortion

(A) Marginal externality

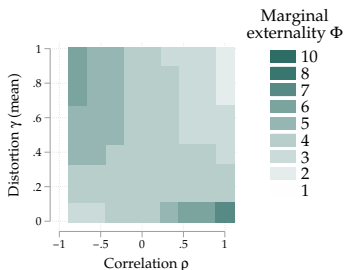
Impact of distortions depends on correlation between WTP and externality



Counterfactual II: Varying correlation and distortion

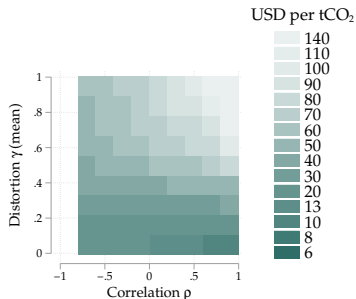
(A) Marginal externality

Impact of distortions depends on correlation between WTP and externality



(B) Abatement cost

Impact of distortions is unambiguous



Result: Demand distortions unambiguously decrease abatement costs in this context, regardless of the correlation coefficient

Selection figure

Today

Background

Theory

RCT Design

Impact

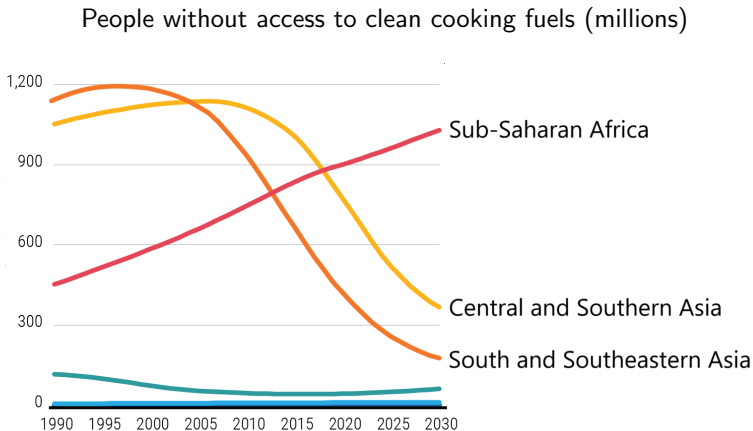
Externality of the marginal adopter

Subsidy expenditure per marginal adoption

Model estimation and counterfactuals

Implications for abatement policy

2.1 billion people lack clean cooking; >1bn in 2030



Source: Joint SDG 7 Report

Electricity access

Where is the lowest portion of the abatement cost curve?

Where is the lowest portion of the abatement cost curve?

Larger marginal cost subsidy

- ▶ Sophisticated credit markets
80% of US cars sold with financing
- ▶ Uncorrelated private benefits and externality
- ▶ Heterogeneous baseline technologies
- ▶ Heterogeneous substitute technologies
- ▶ High intensive margin elasticity

Larger fixed subsidy

- ▶ Larger demand distortions
credit constraints, inattention, myopia
- ▶ Positively correlated private and social benefits, *e.g. fuel usage with proportional prices and emissions*
- ▶ Inelastic intensive margin
e.g. daily routines
- ▶ Uniform baseline technology
- ▶ A single substitute technology

Efficiency of selected green subsidies

Technology	Observable welfare gain per US\$1 subsidy (US\$)	Subsidy spend per tCO ₂ e abated (US\$)	Source
<i>(a) Low- and middle-income contexts</i>			
Electric stoves	\$11	\$13	<i>This paper</i>
Solar lanterns	12	10	Rom et al. (2023)
Energy efficient motors	4.1	— ^a	Chaurey et al. (2025)
Improved charcoal stoves	322 ^b	6	Berkouwer & Dean (2026); Hahn et al. (2024)
<i>(b) High-income contexts</i>			
Electric vehicles	0.4	1,356	Hahn et al. (2024)
Electric vehicles	0.3	1,202	Allcott et al. (2026)
Wind production credits	4.9	46	Hahn et al. (2024)
Residential solar	2.9	90	Hahn et al. (2024)
Appliance rebates	0.2	474	Hahn et al. (2024)
Vehicle retirement	0	876	Hahn et al. (2024)
Hybrid vehicles	0	5,940	Hahn et al. (2024)
Weatherization	0	779	Hahn et al. (2024)

Notes: ^aFree dissemination to non-adopters; subsidy spend per abatement not computable.

^bReflects low cost (US\$40), large reductions (2.1 tCO₂e/yr) and savings, and long lifetime.

Conclusion

- ▶ In theory, demand distortions increase the efficacy of fixed cost subsidies for green technologies

Findings from an RCT in Nakuru, Kenya:

1. Induction stoves abate 2.5 tCO₂e per year
2. Marginal cost subsidies have zero extensive and intensive margin impacts
3. Stove subsidies cost \$13 per tCO₂e abated
4. Demand distortions decrease abatement cost by 41%, by:
 - ▶ Increasing the marginal adopter's positive externality by 19%
 - ▶ Reducing subsidy cost per marginal adopter by 30%
5. Without demand distortions, abatement cost would be \$137 per tCO₂e
6. Lower portions of the abatement cost curve could be in contexts with larger demand distortions

“First best: solve credit markets” ?

Moral hazard and adverse selection persistently undermine credit markets:

- ▶ **Informality** $\Rightarrow$ 1.3 billion adults lack a bank or mobile account World Bank (2025)
- ▶ **Imperfect property rights** $\Rightarrow$ difficulty offering or recovering collateral
- ▶ **Institutional constraints** $\Rightarrow$ lack of insurance (e.g. FDIC), riskier savings $\Rightarrow$ lack of nationwide credit bureaus (e.g. FICO), asymmetric information about borrower quality
- ▶ **Frictions in legal systems** $\Rightarrow$ poor debt recovery in case of default
- ▶ **Relational economies** $\Rightarrow$ non-credible threat of punishment

Banks respond by increasing the pecuniary and non-pecuniary costs of credit:

- ▶ Higher interest rates, shorter repayment periods, shorter grace periods, other-asset collateral, strict eligibility requirements, social penalties
Quinn & Woodruff (2019); Banerjee & Duflo (2005, 2014); Field et al. (2013); Kaboski & Townsend (2012); Gertler, Green, Wolfram (2024); Jack et al. (2023); Fafchamps et al. (2014); Itskhoki & Moll (2019)

High LMIC interest rates increase subsidy efficiency

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Source	Year	Country	Loan type	Monthly rate
Karlan, Mullainathan, Roth	2019	India	Moneylenders	141% ^a
Field, Pande	2008	India	Microfinance	44%
Fink, Jack, Masiye	2020	Malawi	Informal, stated	30–40%
Karlan, Mullainathan, Roth	2019	Philippines	Moneylenders	13%
Karlan, Zinman	2008	South Africa	Formal	8–12%
Mckenzie, Woodruff	2008	Mexico	Micro-lenders	10%
Angelucci, Karlan, Zinman ^b	2015	Mexico	Micro-lending	9%
Egger et al	2022	Kenya	Both	6%
Aleem	1990	Pakistan	Moneylenders	5%
Banerjee, Duflo	2007	India	Both	3–4%
Riley	2024	Uganda	Microfinance	3%
Mckenzie, Woodruff	2004	Sri Lanka	Informal	3%
Banerjee, Duflo	2004	India	Both	2.5–5%
Fafchamps	2000	Kenya, Zimbabwe	Both	2.5–5%
Attanasio et al ^b	2015	Mongolia	Microcredit	2%
Banerjee, Duflo, Glennerster, Kinnan ^b	2015	India	Microfinance	2%
Tarozzi, Desai, Johnson ^b	2015	Ethiopia	Microfinance	1%
Crepon, Devoto, Duflo, Pariente ^b	2015	Morocco	Microfinance	1%

^a4.7% per day; ^bPart of the *AEJ: Applied Economics* special issue on microcredit

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Even higher unrealized **returns** across settings

Source	Year	Country	Monthly returns
Banerjee, Duflo	2012	India	89%
Berkouwer, Dean	2022	Kenya	82%
Mckenzie, Woodruff	2008	Mexico	20-33%
Anagol, Udry	2006	Ghana	20-25%
Crepon, Devoto, Duflo, Pariente	2015	Morocco	20%
Mckenzie, Woodruff	2004	Sri Lanka	15%
Field, Pande, Papp, Rigol	2013	India	11%
Kremer, Lee, Robinson, Rostapshova	2013	Kenya	8%
Banerjee, Duflo	2004	India	8%
Dupas, Robinson	2013	Kenya	6%
De Mel, McKenzie, Woodruff	2008	Pakistan	5%
Anagol, Udry	2006	Ghana	5%
Fink, Jack, Masiye	2020	Malawi	>4.5%
Banerjee & Duflo	2005	Growth theory: heterogeneous returns to the same factor	

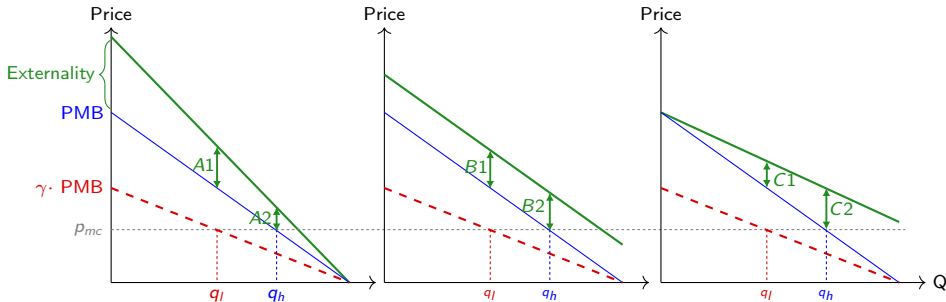
How correlation generates beneficial vs. adverse selection

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[Back to Simulation](#)

(a) Positive correlation
($\rho = 1$)

(b) No correlation
($\rho = 0$)

(c) Negative correlation
($\rho = -1$)



- ▶ With no distortion: positive correlation causes *adverse* selection ($A2 < C2$)
- ▶ With distortion: positive correlation causes *beneficial* selection ($A1 > C1$)
- ▶ Distortions increase marginal externality when $\rho > 0$ ($A1 > A2$), decrease it when $\rho < 0$ ($C1 < C2$)

Full model setup

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Building on Allcott, Mullainathan, Taubinsky (2014)

- ▶ Each individual has inelastic energy services demand θ_i (e.g. cooking, driving)
- ▶ At $t = 1$ they decide whether to buy the greener technology, $d_{1i} = \{0, 1\}$
- ▶ At $t = 2$ they choose what fraction $d_{2i} \in [0, 1]$ of θ_i to consume using the greener substitute
- ▶ Usage benefit: $f(d_{2i}, \pi', \theta_i, \alpha_i) = d_{2i}\theta_i\pi' + \mu(d_{2i}\theta_i - \alpha_i)$
 - ▶ $\mu'' < 0$ (diminishing returns to switching); $\mu(0) = 0$
 - ▶ Optimal usage: $\mu'(x^*) = -\pi'$ where $\pi' = \pi + \tau + s_2$
- ▶ Higher taste for clean technology (α_i) means...
 - ▶ ... larger externality reduction: $\phi_i = (x^* + \alpha_i)\bar{\phi}$
 - ▶ ... higher WTP: $wtp_i = (x^* + \alpha_i)\pi' + \mu(x^*) = a + \alpha_i\pi'$
- ▶ Demand distortion: $\gamma = \frac{1}{1+r}$ where r is the implied interest rate
- ▶ Market demand: $Q(p - s_1) = \sum_i \mathbb{1}[\gamma \cdot wtp_i \geq p - s_1]$

2,134 study participants

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Enrolled at their homes by Busara enumerators, subject to eligibility criteria:

	Mean	25 th	50 th	75 th	N
Number of residents	5.0	4	5	6	2134
Age	35.8	28	35	42	2134
Male (=1)	0.1	0	0	0	2134
Charcoal cooking time (minutes/day)	186.0	140	180	220	2134
Charcoal spending (USD/month)	19.4	14	16	23	2134
LPG spending (USD/month)	9.1	8	8	9	1228
Respondent income (USD/month)	101.5	46	69	116	2127
Household income (USD/month)	246.9	116	196	300	1815
Hours of power outage (30d)	13.9	1	4	12	2134
Hours of voltage fluctuation (30d)	2.0	0	0	1	2134

Balance on baseline characteristics

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	Sample Mean	Credit Treatment	Low BDM Price (i30)	Electricity Treatment	N
	(1)	(2)	(3)	(4)	
Stratification variable: Charcoal spending (USD/month)	19.43 [6.20]	0.23 (0.27)	0.02 (0.27)	-0.04 (0.28)	2134
Stratification variable: Use lpg/gas/ethanol for cooking (=1)	0.58 [0.49]	0.00 (0.02)	0.00 (0.02)	0.00 (0.02)	2134
Stratification variable: Electricity top-up spending (USD, 5 weeks)	4.68 [4.32]	0.39** (0.19)	-0.11 (0.19)	-0.20 (0.20)	2134
Household size	4.95 [1.65]	0.01 (0.07)	0.08 (0.07)	0.03 (0.08)	2134
Age	35.84 [10.15]	0.06 (0.44)	-0.26 (0.44)	-0.18 (0.47)	2134
Male respondent	0.10 [0.30]	0.00 (0.01)	0.02 (0.01)	0.01 (0.01)	2134
Charcoal cooking time (minutes/day)	186.00 [71.08]	-2.02 (3.08)	-3.09 (3.08)	3.00 (3.26)	2134
LPG/ethanol spending (USD/month)	5.23 [10.16]	0.54 (0.44)	-0.49 (0.44)	-0.06 (0.47)	2128
Respondent income (USD/month)	101.44 [115.35]	2.33 (5.00)	1.09 (5.01)	-2.86 (5.29)	2127
Household income (USD/month)	246.99 [204.86]	16.12* (9.61)	16.38* (9.62)	-12.88 (10.18)	1815
Blackout hours (last 30 days)	13.93 [27.49]	2.24* (1.19)	0.50 (1.19)	-0.44 (1.26)	2134
Voltage fluctuation hours (last 30 days)	2.04 [10.88]	-0.70 (0.47)	-0.44 (0.47)	0.20 (0.50)	2134
Joint F-Test		0.09	0.09	0.21	

Attrition by treatment and adoption

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2,134 respondents completed the main survey; 2,081 completed the endline survey.

Predictors of attrition		Reason for attrition	
	Control Mean	Attrited (Visit 3)	Reason
Credit treatment (=1)	0.030	-0.010 (0.007)	Completed survey
BDM price $\geq$ 30 USD	0.027	-0.005 (0.007)	Unavailable
Electricity treatment $\geq$ 0	0.021	0.006 (0.007)	Withdrew from study
Bought induction stove (=1)	0.031	-0.022*** (0.007)	Relocated outside survey team reach
			Hospitalized
			Total
Joint F-Test p-Value		0.00	
Sample Mean		0.02	

- ▶ Attrition balanced across treatments
- ▶ Adopters 2.2pp less likely to attrit; 68% of attriters relocated outside study area

Information about marginal costs of electricity

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No information (15%)

NI NINI NAWENZA KUPIKA KUTUMIA ECOA YANGU?

 UGALI	 ecoQ stoves for life	GITHERI 
 SUKUMA WIKI		TEA 
 RICE		MBOGA 
 BEANS		CHAPATI 
 FISH		BEEF STEW 

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Information about marginal costs of electricity

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Information (85%)

ECOQ YANGU ITATUMIA UNIT/TOKENS NGAPI ZA STIMA KUPIKA?

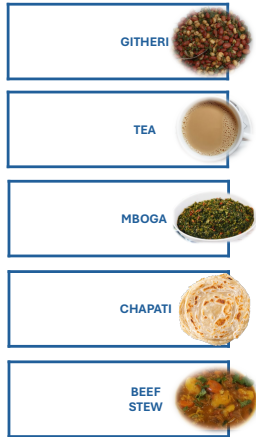


Information about marginal costs of charcoal

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No information (15%)

WHAT CAN I USE MY CHARCOAL JIKO TO COOK?



Information about marginal costs of charcoal

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Information (85%)

HOW MANY TINS OF CHARCOAL DOES MY JIKO USE TO COOK?

ESTIMATES FOR A HOUSEHOLD OF 4 PEOPLE



UGALI



15 KSH



SUKUMA WIKI



10 KSH



RICE



30 KSH



BEANS



45 KSH



FISH



10 KSH



70 KSH



GITHERI



15 KSH



TEA



10 KSH



MBOGA



20 KSH



CHAPATI



25 KSH



BEEF STEW



Converting energy usage to CO₂e emissions

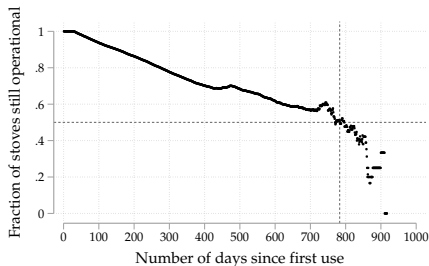
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- ▶ **Charcoal:** (MoFuSS; Bailis et al.; Gill-Wiehl et al.; Floess et al.)
 - ▶ Calorific value: 0.027 TJ per tonne of charcoal
 - ▶ Emissions factor: 161 tCO₂e/TJ combining CO₂, CH₄, N₂O
 - ▶ Fraction non-renewable biomass (fNRB) 38%
 - ▶ 4.3 kgCO₂e per kg charcoal (mean: 3.1 kg charcoal per day, 4.9 tCO₂e per year)
 - ▶ Repeated market price survey ⇒ \$0.21 per kg charcoal
- ▶ **Electricity:** 85% of Kenya's grid is renewable (Kenya Power, 2023)
 - ▶ Average grid emissions intensity: 154 gCO₂e per kWh
 - ▶ Robustness with U.S. grid intensity: 366 gCO₂e per kWh
- ▶ **LPG:** Standard conversion from expenditures to cylinders to CO₂e
- ▶ **Firewood:** 17% of households report using firewood; not tracked
⇒ abatement estimates are conservative

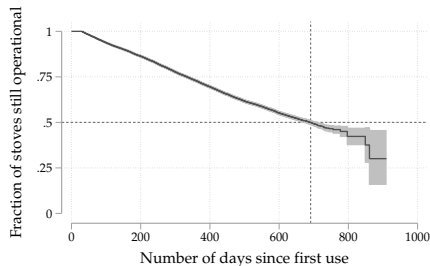
Stove lifespan: 22,304 stoves, Cox model

[Back to Energy spending](#)

(a) Empirical survival



(b) Estimated hazard function



- ▶ 22,304 stoves sold on or before 31 Dec 2024 (real-time usage data)
- ▶ Time variable: days between first and last observed usage
- ▶ Breakage: last usage >3 months before end of dataset
- ▶ Cox proportional hazard model $\Rightarrow$ median lifespan 1.9 years

Other impacts: foods cooked

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	Control Mean	Treatment Effect	N
Number of unique dishes	4.42 [1.09]	0.06 (0.14)	2070
Tea/Coffee/Milk	0.95 [0.22]	0.00 (0.03)	2070
Ugali/Mukimo/Matoke	0.90 [0.30]	-0.12*** (0.04)	2070
Rice/lentils	0.55 [0.50]	0.11* (0.06)	2070
Leafy Greens	0.80 [0.40]	-0.06 (0.05)	2070
Beans/Corn	0.32 [0.47]	0.02 (0.06)	2070
Potato	0.30 [0.46]	0.03 (0.06)	2070
Chapati/Pancakes	0.14 [0.35]	0.06 (0.04)	2070
Fish	0.14 [0.35]	-0.02 (0.04)	2070
Meat Stew	0.09 [0.29]	0.01 (0.04)	2070
Eggs	0.08 [0.28]	0.03 (0.04)	2070
Porridge	0.11 [0.31]	-0.04 (0.04)	2070

Other impacts: health outcomes

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	Control Mean	Treatment Effect	N
Persistent cough (=1)	0.22 [0.41]	-0.14*** (0.05)	2081
Breathless at night (=1)	0.16 [0.37]	-0.09** (0.04)	2081
Child: Persistent cough (=1)	0.40 [0.49]	-0.14* (0.08)	1324
Child: Breathless at night (=1)	0.15 [0.36]	-0.09 (0.06)	1324

Self-reported, for adults and children (HHs with child <5).

- Welfare-relevant, but already incorporated into WTP and usage — not separately monetized

Subsidy cost per tCO₂e under alternative assumptions

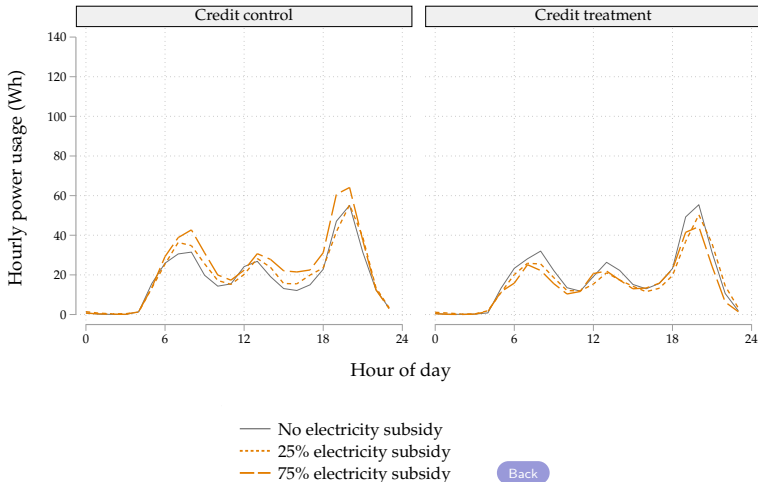
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Assumption		Cost per tCO ₂ e
	Benchmark	US\$13
(1)	One-year durability	US\$25
(2)	Three-year durability	US\$8
(3)	Fraction non-renewable biomass (fNRB) of 10%	US\$16
(4)	Fraction non-renewable biomass (fNRB) of 100%	US\$10
(5)	US grid emissions intensity	US\$13

- ▶ fNRB matters less because around half of charcoal emissions are non-combustion
- ▶ Grid intensity matters little: stove adds ~0.05 tCO₂e at U.S. grid vs. 2.5 tCO₂e saved

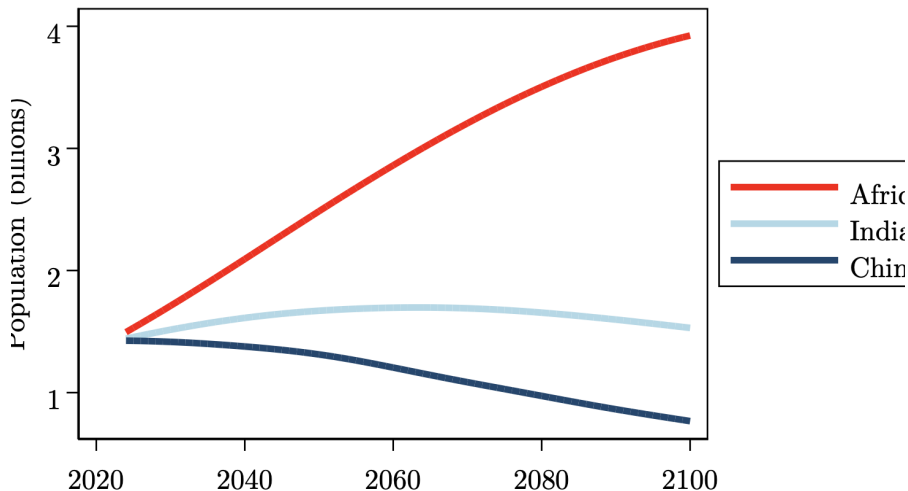
Intensive margin is very price inelastic (by credit group)

- Demand elasticity of stove usage to electricity price: -0.04



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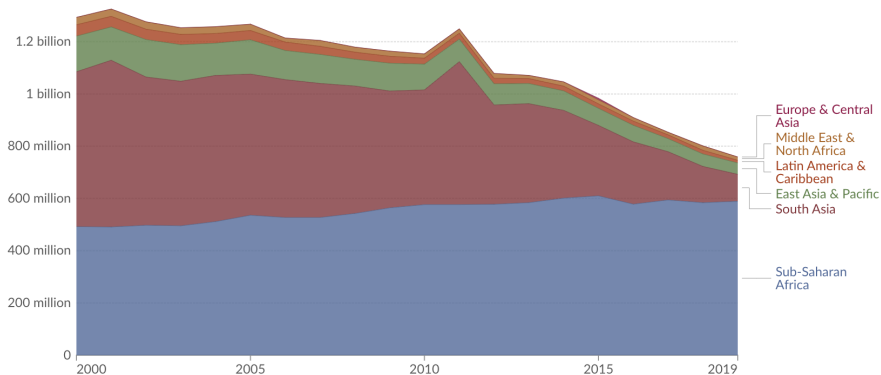
Population in China, India, Africa



Source: UN, World Population Prospects (2024), medium scenario

>700 million people without electricity

Many use kerosene (paraffin) for lighting



Source: World Bank

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